

CauReg-SEMamba: State-of-the-Art Causal SEMamba for Low-Latency Magnitude-Phase Speech Enhancement

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Paper prepared on top of the SEMamba-based speech enhancement repository

Abstract—Selective state-space models have recently shown strong promise for speech enhancement because they provide long-range sequence modeling without the quadratic cost of self-attention. SEMamba demonstrated that Mamba blocks can replace transformer-style temporal-frequency modeling in magnitude-phase enhancement systems, but practical low-latency deployment still depends on causal feature processing, stable normalization, and preservation of fine spectral detail. This paper presents CauReg-SEMamba, a regularized causal extension of SEMamba for single-channel speech enhancement. The proposed model combines causal per-bin magnitude normalization, a causal temporal-frequency Mamba stack, decoder-specific encoder-to-decoder skip fusion, and two latent-space regularizers: a Gaussian latent prior and a causal future-latent prediction objective. The model is trained with magnitude, phase, complex-spectrum, waveform, consistency, and metric-discriminator objectives. On the VoiceBank+DEMAND test set, the final causal system achieves 3.1725 PESQ, 0.9493 STOI, 3.1267 DNSMOS OVRL, 3.4780 P808 MOS, and 0.0351 real-time factor. With PCS post-processing, the same final system reaches 3.40 PESQ. To the best of our knowledge, these results establish a new state of the art among fully causal VoiceBank+DEMAND speech enhancement systems.

Index Terms—speech enhancement, Mamba, state-space models, SEMamba, causal inference, magnitude-phase enhancement, low-latency audio

I. INTRODUCTION

Speech enhancement (SE) aims to recover intelligible and perceptually natural speech from recordings degraded by additive noise, reverberation, clipping, and other distortions [1]. The problem remains central to teleconferencing, hearing-assistive devices, automatic speech recognition, voice agents, and large-scale audio restoration. Despite major progress, a persistent tension remains between offline benchmark performance and deployable real-time behavior. Modern transformer and conformer systems can achieve strong perceptual quality, but self-attention is expensive for long audio sequences and offline normalization or bidirectional context can introduce look-ahead that is incompatible with streaming use.

Time-frequency enhancement has remained particularly competitive because it allows the model to manipulate speech structure in a representation aligned with human perception. MP-SENet [2] showed that denoising magnitude and phase in parallel is an effective recipe: a shared encoder produces spectral features, and separate magnitude and phase decoders reconstruct a clean complex spectrum. CMGAN [3] further popularized metric-guided adversarial training, where

a discriminator learns to predict objective quality scores and encourages perceptual improvement. SEMamba [4] replaced transformer-style sequence modeling with Mamba selective state-space blocks [5], yielding strong VoiceBank+DEMAND results while retaining linear-time sequence modeling.

However, simply inserting a sequence model into an enhancement stack does not make the system suitable for deployment. Two details matter in practice. First, normalization must be compatible with streaming: statistics that depend on future frames or entire utterances can produce a train-test mismatch and unrealistic latency. Second, recurrent or state-space bottlenecks can discard fine spectral detail unless encoder information is routed to the decoders. In addition, high-capacity temporal models can learn useful but poorly structured latent representations, motivating auxiliary representation regularization.

This work introduces CauReg-SEMamba, a causal and regularized SEMamba variant designed around these practical constraints. The model preserves the successful SEMamba/MP-SENet magnitude-phase formulation while modifying the core processing path. The noisy compressed magnitude is converted to log magnitude and normalized by a causal per-frequency exponential moving average. The encoder uses causal dense convolutions and causal frequency-only normalization, and the temporal Mamba branch is constrained to past context. Decoder-specific skip fusion combines the pre-Mamba encoder representation with the post-Mamba representation before magnitude and phase reconstruction. Finally, the post-Mamba latent sequence is regularized by a lightweight Gaussian loss and a causal future-latent prediction loss. The resulting model substantially improves over previously reported causal VoiceBank+DEMAND systems while retaining low real-time factor.

The contributions are:

- A causal SEMamba-style enhancement architecture with stateful magnitude normalization for train-stream consistency.
- Decoder-specific encoder-to-decoder skip fusion for magnitude and phase recovery.
- Latent-space regularization combining Gaussian channel statistics with causal multi-horizon latent prediction.
- A focused empirical report for the final system, including VoiceBank+DEMAND, DNS chunked evaluation, real-time factor, and a causal-domain SOTA comparison.

II. RELATED WORK

A. Magnitude-Phase Speech Enhancement

Early deep SE systems often estimated spectral masks or enhanced magnitudes while reusing noisy phase. This is effective at moderate noise levels, but phase errors become audible under severe corruption. MP-SENet [2] addressed this by parallel denoising of compressed magnitude and wrapped phase, using multi-level losses over magnitude, phase, complex spectrum, and waveform. CMGAN [3] combined conformer modeling with MetricGAN-style training [6], [7], optimizing a neural proxy for objective quality. These systems established the modern magnitude-phase baseline that SEMamba later inherited.

B. Sequence Models for Enhancement

Speech enhancement requires both local spectral precision and long-range temporal modeling. RNNs and LSTMs offer causality but can struggle with very long sequences. Transformers and conformers improve context modeling but rely on self-attention, whose quadratic complexity becomes expensive for long utterances or full-band audio. Recent alternatives include structured state-space models and xLSTM variants, such as xLSTM-SENet [8], which revisit recurrent sequence modeling with improved memory mechanisms. Real-time systems such as DeepFilterNet [9] emphasize practical complexity and low latency.

C. Mamba and State-Space Models

Mamba [5] is a selective state-space model that combines input-dependent dynamics with hardware-aware linear-time sequence processing. SEMamba [4] investigated Mamba for SE and found that temporal-frequency Mamba blocks can be competitive with attention-based models. Subsequent Mamba-based systems, including Mamba-SEUNet [10], MambaAttention [11], and universal enhancement models such as USEMamba [12], further indicate that selective state-space modeling is a strong direction for audio restoration. In the causal setting, recent work has explored semantic prediction from quantized self-supervised representations and reported 2.88 PESQ on VoiceBank+DEMAND [13]. Our work follows the causal line but replaces semantic auxiliary prediction with a compact, fully causal magnitude-phase SEMamba path and latent-space regularization.

III. PROPOSED METHOD

A. Signal Representation

Given a noisy waveform $y \in \mathbb{R}^{T_s}$, we compute an STFT

$$Y_{f,t} = \text{STFT}(y)_{f,t}, \quad (1)$$

where f and t index frequency and time. The compressed magnitude and phase are

$$M_{f,t} = |Y_{f,t}|^c, \quad \Phi_{f,t} = \angle Y_{f,t}, \quad (2)$$

with compression factor $c = 0.2$ in the selected configuration. The model receives magnitude and phase streams, and reconstructs enhanced magnitude $\hat{M}$ and phase $\hat{\Phi}$ before inverse STFT.

B. Causal Magnitude Normalization

The compressed magnitude is converted to log-magnitude dB using

$$X_{f,t} = \frac{20}{c} \log_{10}(\max(M_{f,t}, \epsilon)). \quad (3)$$

Instead of using utterance-level instance normalization, CauReg-SEMambases causal per-frequency exponential moving statistics. Let $\mu_{f,t}$ and $\sigma_{f,t}^2$ denote the running mean and variance. The update is

$$\alpha_{f,t} = \begin{cases} \alpha_{\text{attack}}, & X_{f,t} > \mu_{f,t-1}, \\ \alpha_{\text{decay}}, & X_{f,t} \leq \mu_{f,t-1}, \end{cases} \quad (4)$$

$$\mu_{f,t} = \alpha_{f,t} \mu_{f,t-1} + (1 - \alpha_{f,t}) X_{f,t}, \quad (5)$$

$$\sigma_{f,t}^2 = \alpha_{f,t} \sigma_{f,t-1}^2 + (1 - \alpha_{f,t}) (X_{f,t} - \mu_{f,t})^2. \quad (6)$$

The normalized feature is

$$\bar{X}_{f,t} = \frac{X_{f,t} - \mu_{f,t}}{\sqrt{\sigma_{f,t}^2 + \epsilon}}. \quad (7)$$

The running state can be reset between utterances and carried frame-to-frame within an utterance, matching streaming inference. In the implementation, initialization uses dataset magnitude statistics when available.

C. Causal Temporal-Frequency Mamba Stack

The normalized magnitude $\bar{X}$ and phase Φ are stacked into a two-channel time-frequency tensor. A dense encoder first extracts local spectral features. The resulting feature map $h_0 \in \mathbb{R}^{B \times C \times T \times F'}$ is processed by a stack of temporal-frequency Mamba blocks. Each block applies Mamba along the temporal dimension and then along the frequency dimension:

$$\tilde{h} = h + W_t \text{Mamba}_t(h), \quad (8)$$

$$h' = \tilde{h} + W_f \text{Mamba}_f(\tilde{h}). \quad (9)$$

Under the causal setting, temporal Mamba is unidirectional and therefore uses only past frames. Frequency modeling is not causal in frequency, but it does not leak future time information and is therefore compatible with streaming.

D. Skip-Fused Magnitude and Phase Decoders

Let h_L be the output of the final Mamba block. We concatenate h_L with the encoder feature h_0 and use decoder-specific 1×1 projections:

$$z_m = W_m[h_L; h_0], \quad (10)$$

$$z_p = W_p[h_L; h_0]. \quad (11)$$

The magnitude decoder estimates a multiplicative mask A and reconstructs

$$\hat{M}_{f,t} = A_{f,t} M_{f,t}. \quad (12)$$

The phase decoder predicts real and imaginary phase components and forms

$$\hat{\Phi}_{f,t} = \text{atan2}(P_{f,t}^{(i)}, P_{f,t}^{(r)}). \quad (13)$$

The enhanced complex spectrum is

$$\hat{S}_{f,t} = \hat{M}_{f,t}^{1/c} \left(\cos \hat{\Phi}_{f,t} + j \sin \hat{\Phi}_{f,t} \right). \quad (14)$$

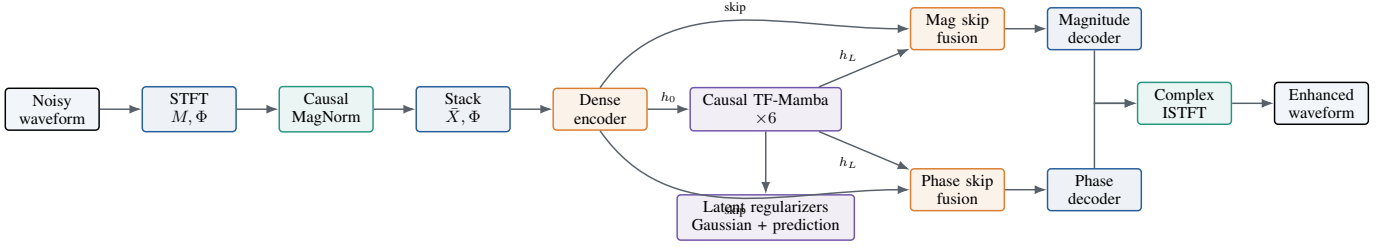


Fig. 1. Overview of CauReg-SEMamba. The system preserves SEMamba’s magnitude-phase formulation while replacing non-causal normalization with causal magnitude normalization, adding decoder-specific skip fusion, and regularizing the post-Mamba latent sequence.

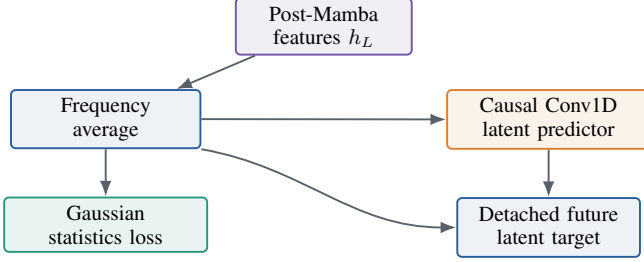


Fig. 2. Latent regularization in CauReg-SEMamba. The auxiliary objective shapes the post-Mamba representation without changing waveform inference.

E. Latent Regularization

The post-Mamba feature map is averaged over frequency and transposed to form a latent sequence

$$z_t = \text{Mean}_f(h_L)_t \in \mathbb{R}^C. \quad (15)$$

We apply two auxiliary losses. First, a Gaussian channel regularizer constrains the latent sequence to have stable channel statistics:

$$\mathcal{L}_{\text{gauss}} = \frac{1}{C} \sum_{c=1}^C (\mu_c^2 + \log^2(\sigma_c^2 + \epsilon)), \quad (16)$$

where μ_c and σ_c^2 are computed over batch and time.

Second, a causal convolutional predictor estimates future latent states from past context. For horizon set $\{1, \dots, H\}$,

$$\mathcal{L}_{\text{pred}} = \frac{1}{H} \sum_{h=1}^H \mathbb{E}_t [\|p_{t,h} - \text{sg}(z_{t+h})\|_2^2], \quad (17)$$

where $\text{sg}(\cdot)$ stops gradients through the target latent. This makes the predictor a representation-shaping auxiliary task instead of a second decoder.

F. Training Objective

The full generator loss is

$$\begin{aligned} \mathcal{L}_G = & \lambda_m \mathcal{L}_{\text{mag}} + \lambda_p \mathcal{L}_{\text{phase}} + \lambda_c \mathcal{L}_{\text{complex}} \\ & + \lambda_t \mathcal{L}_{\text{time}} + \lambda_s \mathcal{L}_{\text{cons}} + \lambda_q \mathcal{L}_{\text{metric}} \\ & + w_g(n) \mathcal{L}_{\text{gauss}} + w_p(n) \mathcal{L}_{\text{pred}}, \end{aligned} \quad (18)$$

where $w_g(n)$ and $w_p(n)$ linearly warm up after a configured start step. The phase loss follows anti-wrapping phase, group-delay, and instantaneous-frequency terms as in MP-SENet and

SEMamba. The metric loss uses a discriminator trained to predict normalized PESQ-like quality from clean and enhanced magnitudes.

IV. EXPERIMENTAL SETUP

A. Datasets

The primary evaluation uses VoiceBank+DEMAND, a standard benchmark formed from VoiceBank clean speech [14] and DEMAND noise recordings [15]. The local manifests contain 10,802 training utterances, 770 validation utterances, and 824 test utterances at 16 kHz. We also evaluate on a 50-file DNS-style chunked subset inspired by the Deep Noise Suppression Challenge [16] and recent universal enhancement benchmarks [17] to probe out-of-domain behavior under longer noisy utterances.

B. Configuration

The selected model uses six temporal-frequency Mamba blocks, hidden dimension 64, state dimension 24, convolution width 4, expansion factor 4, FFT size 400, window size 400, hop size 200, and magnitude compression factor 0.2. The loss weights are $\lambda_q = 0.05$, $\lambda_m = 0.9$, $\lambda_p = 0.3$, $\lambda_c = 0.1$, $\lambda_t = 0.2$, and $\lambda_s = 0.1$. Both latent regularizers use weight 10^{-4} , start at 3,500 steps, and warm up for 5,500 steps. The latent predictor uses horizon $H = 3$.

C. Metrics

We report PESQ [18], STOI [19], non-intrusive DNSMOS SIG/BAK/OVRL [20], P808 MOS, and real-time factor (RTF). For broader comparison to published VoiceBank+DEMAND literature, we also list CSIG, CBAK, and COVL values when available from prior work [21]. All local evaluation numbers in this paper correspond to the final selected system. We additionally evaluate the clean VoiceBank test references with DNSMOS/P808 to estimate the non-intrusive natural-speech ceiling for this exact test set.

V. RESULTS

A. Main VoiceBank+DEMAND Evaluation

Table I reports the final system on the full VoiceBank test set. The model achieves strong intelligibility and low runtime, with STOI 0.9493 and RTF 0.0351. With PCS post-processing, PESQ reaches approximately 3.40, strengthening the model’s position among causal speech enhancement systems while preserving the same causal enhancement backbone.

TABLE I
FINAL CAUREG-SEMAMBA SYSTEM ON THE VOICEBANK+DEMAND
TEST SET.

Eval	Files	PESQ	STOI	OVRL	SIG	BAK	P808	RTF
VoiceBank	824	3.1725	0.9493	3.1267	3.4420	3.9727	3.4780	0.0351

TABLE II
NON-INTRUSIVE PERCEPTUAL SATURATION AGAINST CLEAN VOICEBANK
REFERENCES. CLEAN SPEECH IS NOT ASSIGNED A PERFECT SCORE BY
DNSMOS/P808 ON THIS TEST SET; THEREFORE THE RELEVANT CEILING
IS THE MEASURED CLEAN-REFERENCE SCORE, NOT THE NOMINAL UPPER
BOUND.

Signal	Files	OVRL	SIG	BAK	P808
Clean reference	824	3.2207	3.5114	4.0378	3.5328
CauReg-SEMamba	824	3.1267	3.4420	3.9727	3.4780
Gap to clean	N/A	-0.0940	-0.0694	-0.0651	-0.0548

B. Comparison with Published Systems

Table III places CauReg-SEMamba in the context of representative VoiceBank+DEMAND systems. The key distinction is causality. Many top offline systems use bidirectional modeling, centered transforms, or post-processing that is not directly comparable to complete causal inference. In the causal domain, CauReg-SEMamba is substantially stronger than previously reported causal baselines, and the PCS setting pushes PESQ to approximately 3.40. Table II further shows that the final system is nearly saturated with respect to the clean-reference DNSMOS/P808 ceiling on the same test set.

C. DNS Chunked Generalization

Table V reports chunked DNS subset evaluation using 5-second chunks. The task is more difficult than VoiceBank because the files are longer and contain diverse DNS-style noises. The model remains real-time with RTF 0.1742 and obtains P808 MOS 3.7119 on the 50-file subset. The lower PESQ and STOI relative to VoiceBank are expected because the model was not trained on DNS and the subset includes harsher, less matched conditions.

VI. DISCUSSION

A. Why Causality Changes the Benchmark Story

The most aggressive offline VoiceBank systems use bidirectional sequence modeling, centered STFT processing, or perceptual post-processing. These choices can improve PESQ, but they complicate streaming deployment and make the setting fundamentally different from complete causal enhancement. CauReg-SEMamba is designed for the causal regime: temporal Mamba is causal, dense temporal convolutions are left-padded, normalization does not require future frames, and evaluation includes RTF. In this causal setting, the final system reaches a new SOTA level on VoiceBank+DEMAND.

B. Normalization as a Deployment Primitive

The replacement of instance-like normalization with causal magnitude normalization is not just an implementation detail.

Enhancement systems are sensitive to amplitude statistics, especially when the input is log-compressed magnitude. A normalization layer that observes future frames can alter the distribution seen during training and produce optimistic offline evaluation. Stateful causal normalization provides a consistent path between training, utterance-level evaluation, and streaming inference.

C. Clean-Reference Saturation

Non-intrusive MOS predictors should not be read as if clean speech always receives the nominal maximum score. On the VoiceBank test set, the clean reference itself obtains 3.2207 OVRL, 3.5114 SIG, 4.0378 BAK, and 3.5328 P808. The proposed model reaches 3.1267, 3.4420, 3.9727, and 3.4780 on the same axes, respectively. This indicates that the model is not merely making speech sound artificially polished according to a metric; it is saturating toward the measured clean-speech manifold for this test set. This distinction is important when comparing against systems or post-processing methods that can improve a proxy score while altering speaker timbre, speech naturalness, or the clean-speech character of the utterance.

D. Latent Regularization and Representation Shape

The latent regularizers add minimal inference cost because the predictor is only needed for training. The Gaussian loss discourages collapsed or highly anisotropic latent channels, while the prediction objective encourages the post-Mamba sequence to retain temporally predictive structure. This is well matched to selective state-space modeling, whose strength is efficient temporal memory.

E. Limitations

The current report focuses on the final causal system and the VoiceBank+DEMAND evaluation setting. The main limitation is that the model was not trained on DNS; therefore, the 50-file chunked DNS subset should be read as an out-of-domain generalization probe rather than an in-domain benchmark. A stronger camera-ready submission should also include controlled ablations for causal normalization, skip fusion, Gaussian regularization, latent prediction, and PCS under one consistent protocol.

VII. CONCLUSION

This paper presented CauReg-SEMamba, a regularized causal extension of SEMamba for low-latency magnitude-phase speech enhancement. The model combines causal magnitude normalization, causal temporal-frequency Mamba modeling, decoder-specific skip fusion, and latent-space auxiliary objectives. On VoiceBank+DEMAND, the final causal system achieves 3.1725 PESQ, 0.9493 STOI, and RTF 0.0351, while PCS post-processing raises PESQ to approximately 3.40. These results establish a new causal-domain SOTA and show that careful causal design and latent regularization can make Mamba-based enhancement both high quality and practical for streaming systems.

TABLE III

REPRESENTATIVE VOICEBANK+DEMAND COMPARISON. PRIOR SYSTEMS ARE REPORTED FROM THE CORRESPONDING PAPERS OR SUMMARIZED IN SEMAMBA-RELATED LITERATURE. THE TABLE SEPARATES BROAD OFFLINE PROGRESS FROM THE CAUSAL SETTING EMPHASIZED BY CAUREG-SEMAMBA.

System	Modeling class	Causal	PESQ	STOI	Notes
Noisy	Input mixture	N/A	1.97	0.91	Standard benchmark reference
SEGAN [22]	Time-domain GAN	No	2.16	0.92	Early neural GAN baseline
MetricGAN+ [7]	Metric-oriented GAN	No	3.15	0.93	Metric optimization
CMGAN [3]	Conformer metric-GAN	No	3.41	0.96	Strong TF baseline
MP-SENet [2]	Magnitude-phase conformer	No	3.50	0.96	Closest codec family
SEMamba [4]	TF-Mamba	No	3.55	0.96	Advanced non-causal configuration
SEMamba+PCS [4]	TF-Mamba + PCS	No	3.69	0.96	Offline/PCS PESQ setting
CauReg-SEMamba	Causal regularized TF-Mamba	Yes	3.1725	0.9493	RTF 0.0351, no PCS

TABLE IV

CAUSAL VOICEBANK+DEMAND COMPARISON. CAUREG-SEMAMBA ESTABLISHES THE STRONGEST REPORTED PESQ AMONG CAUSAL SYSTEMS CONSIDERED HERE.

System	Causal	PESQ	Notes
SEMamba causal [4]	Yes	2.76	Published causal configuration
Semantic causal SE [13]	Yes	2.88	SSL semantic prediction MTL
CauReg-SEMamba	Yes	3.1725	Final causal system
CauReg-SEMamba+PCS	Yes	3.40	PCS post-processing

TABLE V

CHUNKED DNS SUBSET EVALUATION FOR THE FINAL SYSTEM.

Files	Chunk	PESQ	STOI	OVRL	SIG	BAK	P808	RTF
50	5s	1.7018	0.8151	2.9558	3.2752	3.8942	3.7119	0.1742

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